SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: ITA 0603 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)*

Assessment Tool Weightage: 40%

QUESTIONS: 15 Total Marks: 25

Annexure A

APPLICATION BASED QUESTIONS

Code of Conduct:

I _M.v.Pranav (192312358)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
pranav mv

Annexure A

Short Answer Test

Q1. What is the probability that the selected box was Box C?

Answer

Concept:

This problem is solved using **Bayes' Theorem**, which finds the probability of an event based on prior information.

Formula:

$$P(C|Blue)=\frac{P(Blue|C)\times P(C)}{P(Blue)}$$

Steps:

- Probability of selecting boxes:
- Box A = 1/4
- Box B = 1/2
- Box C = 1/4
- Probability of drawing a blue ball:
- A = 4/12
- B = 7/15
- C = 2/15
- Apply Bayes' theorem using the above probabilities.
- Calculate the posterior probability.

Final Answer:

Probability = 1/6 = 0.167 = 16.7%

Conclusion:

Bayes' theorem helps determine the probability that the selected box was Box C after observing a blue ball.

Q2. What is the probability that the car came from Section C?

Answer

Concept:

Bayes' Theorem is used because the section is selected first and then a blue car is observed.

Formula:

$$P(C|Blue) = \frac{P(Blue|C) \times P(C)}{P(Blue)}$$

Steps:

- Probability of selecting sections:
- Section A = 1/4
- Section B = 1/2
- Section C = 1/4
- Probability of selecting a blue car:
- A = 3/10
- B = 6/15
- C = 2/15
- Substitute values into Bayes' theorem.
- Compute the final probability.

Final Answer:

Probability = 0.10 = 10%

Conclusion:

The probability that the blue car belongs to Section C is **10%**.

Q3. Find FN, Accuracy, Precision, Recall and Specificity.

Answer

Given:

- TP = 210
- TN = 190
- FP = 30
- FN = 150

Formulas

Accuracy

$$= \frac{TP + TN}{TP + TN + FP + FN}$$

Precision

$$= \frac{TP}{TP+FP}$$

Recall

$$= \frac{TP}{TP+FN}$$

Specificity

$$= \frac{TN}{TN+FP}$$

Results

- False Negative (FN) = **150**
- Accuracy = **55.56%**
- Precision = **87.5%**
- Recall = **58.33%**
- Specificity = **86.36%**

Conclusion

- The classifier has **high precision**, meaning positive predictions are reliable.
- Recall is comparatively lower, indicating some positive cases are missed.
- Suitable for classification but recall can be improved.

Q4. Examine the training and validation loss values.

Answer

i) Identify the epoch at which overfitting begins.

Overfitting starts at **Epoch 4** because after this point the **training loss decreases**, while the **validation loss begins to increase**.

ii) Two techniques to reduce overfitting

1. **Early Stopping**

- Stops training before overfitting occurs.
- Improves model performance.

2. **Dropout**

- Randomly removes neurons during training.
- Prevents memorization of training data.

Other methods:

- L2 Regularization
- Data Augmentation

iii) Effect of overfitting

- Model memorizes training data.
- Validation loss increases.
- Generalization ability decreases.
- Performance on unseen data becomes poor.

Conclusion

The model should stop training around **Epoch 4** to obtain the best performance.

Q5. Examine the training and validation accuracy values.

Answer

i) Identify when overfitting starts.

Overfitting starts at **Epoch 5**.
Reason:

- Training accuracy continues increasing.
- Validation accuracy begins decreasing.

ii) Why validation accuracy decreases?

- The model memorizes the training data.
- It cannot generalize to unseen data.
- Prediction accuracy on validation data reduces.
- The gap between training and validation accuracy increases.

iii) Methods to improve validation performance

1.Early Stopping
2.Data Augmentation
3.Dropout
4.Regularization

Conclusion

Q6. A machine learning model shows the following loss values during training.

Answer

i) At which epoch does the model achieve optimal generalization?

The model achieves the **best generalization at Epoch 4** because:

- Validation loss is the lowest.
- Training and validation losses are close.
- The model performs well on unseen data.

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ii) Identify signs of overfitting.

The signs of overfitting are:

- Training loss keeps decreasing.
- Validation loss starts increasing after Epoch 4.
- The gap between training and validation loss becomes larger.
- The model memorizes training data instead of learning patterns.

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iii) Recommend two regularization techniques and explain their impact.

1. Dropout

- Randomly disables neurons during training.
- Reduces overfitting.
- Improves generalization.

2. L2 Regularization

- Adds a penalty to large weights.
- Prevents complex models.
- Improves model stability.

Other techniques

- Early Stopping
- Data Augmentation

Conclusion

The best model is obtained at **Epoch 4**. Regularization techniques improve performance on unseen data.

Q7. Compute Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer

Given

- TP = 119
- FN = 6
- TN = 120
- FP = 5

Formulas

Accuracy

$$\frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$\frac{TP}{TP+FP}$$

Sensitivity (Recall)

$$\frac{TP}{TP+FN}$$

Specificity

$$\frac{TN}{TN+FP}$$

F1-Score

$$\frac{2\times Precision\times Recall}{Precision+Recall}$$

Results

- **Accuracy = 95.6%**
- **Precision = 96.0%**
- **Sensitivity (Recall) = 95.2%**
- **Specificity = 96.0%**
- **F1-Score = 95.6%**

Conclusion

- The AI robot correctly classifies most tires.
- Both precision and recall are high.
- The model is reliable for industrial automation.

Q8. Compute Accuracy, Precision, Recall, Specificity and F1-Score.

Answer

Given

- TP = 138
- FN = 12
- TN = 135
- FP = 15

Formulas

- Accuracy
- Precision
- Recall
- Specificity
- F1-Score

(using the standard classification formulas)

Results

- **Accuracy = 91.0%**
- **Precision = 90.2%**
- **Recall = 92.0%**
- **Specificity = 90.0%**
- **F1-Score = 91.1%**

Conclusion

- The model detects disease accurately.
- High recall means most diseased patients are identified.
- Suitable for medical diagnosis.

Q9. Compute Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer

Given

- TP = 180
- FN = 20

- TN = 170
- FP = 30

Results

- **Accuracy = 87.5%**
- **Precision = 85.7%**
- **Sensitivity (Recall) = 90.0%**
- **Specificity = 85.0%**
- **F1-Score = 87.8%**

Explanation

- The spam filter correctly detects most spam emails.
- Recall is high because many spam emails are identified.
- Precision indicates that most predicted spam emails are actually spam.
- The F1-score shows balanced performance.

Conclusion

The spam filtering model provides good overall classification accuracy and is suitable for email filtering systems.

Q10. Compute Accuracy, Precision, Sensitivity, Specificity and F1-Score.

Answer

Given

- TP = 230
- FN = 20
- TN = 225
- FP = 25

Results

- **Accuracy = 91.0%**
- **Precision = 90.2%**
- **Sensitivity (Recall) = 92.0%**
- **Specificity = 90.0%**
- **F1-Score = 91.1%**

Explanation

- The AI model correctly identifies most defective products.
- High precision reduces false alarms.
- High recall ensures defective products are detected.
- The F1-score indicates balanced performance.

Conclusion

The quality inspection system is reliable and suitable for detecting defective and non-defective products.

Q11. In Shop X, there are 20 bottles of pure oil and 30 bottles of adulterated oil, while in Shop Y, there are 40 bottles of pure oil and 20 bottles of adulterated oil. A bottle is selected at random from one of the shops, with Shop Y being twice as likely to be chosen as Shop X. The selected bottle is found to be adulterated. What is the probability that it came from Shop Y?

Answer

Concept

This problem is solved using **Bayes' Theorem**, which calculates the probability of an event after observing new information.

Formula

$$P(Y|A)=\frac{P(A|Y)\times P(Y)}{P(A)}$$

where:

- $P(Y|A)$ = Probability that the bottle came from Shop Y.
- $P(A|Y)$ = Probability of selecting an adulterated bottle from Shop Y.

Steps

- Probability of selecting the shops:
- Shop X = **1/3**
- Shop Y = **2/3**
- Probability of adulterated bottle:
- Shop X = **30/50 = 0.60**
- Shop Y = **20/60 = 0.333**
- Apply Bayes' theorem.

- Calculate the posterior probability.

Final Answer

Probability = 0.471 = 47.1%

Conclusion

Bayes' Theorem helps determine the probability that the adulterated bottle came from Shop Y after observing the selected bottle.

Q12. A warehouse has two sections. Section A contains defective and non-defective items, while Section B contains defective and non-defective items. One section is chosen randomly, and a defective item is found. Find the probability that it came from Section A.

Answer

Concept

This problem is solved using **Bayes' Theorem**.

Formula

$$P(A|D)=\frac{P(D|A)\times P(A)}{P(D)}$$

where:

- $P(A|D)$ = Probability that the defective item came from Section A.

Steps

- Section A is **three times** more likely to be selected.
- Probability of selecting:
- Section A = **3/4**
- Section B = **1/4**
- Compute the probability of selecting a defective item.
- Apply Bayes' theorem.
- Calculate the posterior probability.

Final Answer

Probability = 0.643 = 64.3%

Conclusion

The defective item is more likely to have come from **Section A** because it has a higher selection probability.

Q13. Find Accuracy, Precision, Sensitivity (Recall) and Specificity.

Answer

Given

- TP = **120**
- TN = **150**
- Test samples = **400**
- FP = **0**
- FN = **80**

Formulas

Accuracy

$$\frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$\frac{TP}{TP+FP}$$

Recall

$$\frac{TP}{TP+FN}$$

Specificity

$$\frac{TN}{TN+FP}$$

Results

- **Accuracy = 67.5%**
- **Precision = 100%**
- **Recall = 60%**

- **Specificity = 75%**

Conclusion

- The classifier predicts positive cases accurately.
- Some positive samples are missed, resulting in lower recall.
- The model is useful but recall can be improved.

Q14. Medical Image Classification using KNN

Answer

Given

- TP = **180**
- TN = **160**
- Test samples = **600**

Results

- **Accuracy = 56.7%**
- **Precision = 100%**
- **Recall = 60%**
- **Specificity = 53.3%**

Explanation

- Precision is very high because there are no false positive predictions.
- Recall is moderate because some positive cases are not detected.
- Specificity is comparatively lower.
- The classifier performs well in identifying positive samples but needs improvement for overall accuracy.

Conclusion

The model has **high precision** but **moderate recall and specificity**, making it suitable for applications where false positives must be minimized.

Q15. A deep learning model is trained for a classification task. Analyze the loss values.

Answer

a) At which epoch does the model achieve the best generalization performance?

The model achieves the **best generalization at Epoch 4** because the validation loss reaches its minimum value.

b) At which point does overfitting begin? Justify your answer.

Overfitting begins at **Epoch 5** because:

- Training loss continues to decrease.
- Validation loss starts increasing.
- The gap between training and validation loss becomes larger.
- The model begins memorizing the training data.

c) Suggest two techniques to reduce overfitting and explain how they help.

1. Dropout

- Randomly deactivates neurons during training.
- Prevents the model from memorizing data.
- Improves generalization.

2. Early Stopping

- Stops training when validation loss begins to increase.
- Prevents unnecessary training.
- Produces a better-performing model.

Other techniques

- L2 Regularization
- Data Augmentation

Conclusion

The model performs best at **Epoch 4**. Applying regularization techniques helps reduce overfitting and improves performance on unseen data.